

Replay-Aware ATSP Benchmark Report

Overview

- Condition count: 4.
- Best final transfer gap: atsp_random_replay.
- Best TSPLIB ATSP holdout gap: atsp_random_replay.
- Best synthetic holdout gap: atsp_random_replay.

Run Metadata

- run_name: run_20260513_180618_n
- started_at_local: 2026-05-13 18:06:18
- finished_at_local: 2026-05-13 18:14:40
- duration_hhmm: 00:08
- duration_seconds: 502.447
- seed_offset: 13000
- replicate_label: n
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Replay	Selection	Compression	Final TSPLIB ATSP Gap	Final Synthetic Gap	Final Transfer Gap	Mean Accepted Novelty	Mean Complexity	Adaptation Efficiency
atsp_no_replay	none	score_only	False	0.183122	0.009004	0.096063	0.884931	0.58	0.00821
atsp_random_replay	random	score_only	False	0.178271	0.008418	0.093345	0.835038	0.78	0.063932
atsp_failure_replay	failure	score_only	False	0.210045	0.036251	0.123148	0.0	0.78	0.0
atsp_failure_replay_compression	failure	novelty_gate	True	0.193715	0.009004	0.10136	0.750877	0.58	0.020222

Condition Notes

atsp_no_replay

- Replay mode: none with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.183122, synthetic 0.009004, combined 0.096063.

- Accepted-epoch count 6, mean accepted code novelty 0.884931, and final complexity 0.58.
- Adaptation efficiency 0.00821 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.183122 across 4 instances; family means: ft=0.322375, ftv=0.33478, p=0.003915, ry=0.071419.
- Panel synthetic_holdout mean gap 0.009004 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.033816, wind_clusters=0.002199.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1903, "family": "ftv", "name": "ftv38", "optimality_gap": 0.243791}, {"best_known_cost": 1286, "cost": 1537, "family": "ftv", "name": "ftv33", "optimality_gap": 0.195179}, {"best_known_cost": 1473, "cost": 1749, "family": "ftv", "name": "ftv35", "optimality_gap": 0.187373}]

atsp_random_replay

- Replay mode: random with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.178271, synthetic 0.008418, combined 0.093345.
- Accepted-epoch count 2, mean accepted code novelty 0.835038, and final complexity 0.78.
- Adaptation efficiency 0.063932 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 2.
- Panel heldout_tsplib mean gap 0.178271 across 4 instances; family means: ft=0.28979, ftv=0.256045, p=0.012278, ry=0.154972.
- Panel synthetic_holdout mean gap 0.008418 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.015345, wind_clusters=0.018328.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 2006, "family": "ftv", "name": "ftv38", "optimality_gap": 0.311111}, {"best_known_cost": 1286, "cost": 1578, "family": "ftv", "name": "ftv33", "optimality_gap": 0.227061}, {"best_known_cost": 1473, "cost": 1765, "family": "ftv", "name": "ftv35", "optimality_gap": 0.198235}]

atsp_failure_replay

- Replay mode: failure with selection mode score_only.
- Final transfer gaps: TSPLIB ATSP 0.210045, synthetic 0.036251, combined 0.123148.
- Accepted-epoch count 1, mean accepted code novelty 0.0, and final complexity 0.78.
- Adaptation efficiency 0.0 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.
- Panel heldout_tsplib mean gap 0.210045 across 4 instances; family means: ft=0.4, ftv=0.286423, p=0.008007, ry=0.14575.
- Panel synthetic_holdout mean gap 0.036251 across 4 instances; family means: clockwise_ring=0.085613, corridor_drift=0.0, hub_spokes=0.059392, wind_clusters=0.0.
- Worst recent training cases: [{"best_known_cost": 1473, "cost": 1841, "family": "ftv", "name": "ftv35", "optimality_gap": 0.24983}, {"best_known_cost": 1530, "cost": 1869, "family": "ftv", "name": "ftv38", "optimality_gap": 0.221569}, {"best_known_cost": 1286, "cost": 1542, "family": "ftv", "name": "ftv33", "optimality_gap": 0.199067}]

atsp_failure_replay_compression

- Replay mode: failure with selection mode novelty_gate.
- Final transfer gaps: TSPLIB ATSP 0.193715, synthetic 0.009004, combined 0.10136.
- Accepted-epoch count 3, mean accepted code novelty 0.750877, and final complexity 0.58.
- Adaptation efficiency 0.020222 and archive sizes {'worst_cases': 4, 'failure_cases': 3, 'adversarial_layouts': 4} / elite 1.

- Panel heldout_tsplib mean gap 0.193715 across 4 instances; family means: ft=0.323244, ftv=0.33478, p=0.004093, ry=0.112744.
- Panel synthetic_holdout mean gap 0.009004 across 4 instances; family means: clockwise_ring=0.0, corridor_drift=0.0, hub_spokes=0.033816, wind_clusters=0.002199.
- Worst recent training cases: [{"best_known_cost": 1530, "cost": 1939, "family": "ftv", "name": "ftv38", "optimality_gap": 0.26732}, {"best_known_cost": 1473, "cost": 1705, "family": "ftv", "name": "ftv35", "optimality_gap": 0.157502}, {"best_known_cost": 1286, "cost": 1478, "family": "ftv", "name": "ftv33", "optimality_gap": 0.1493}]

Judge Appendix

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Summary of ATSP Benchmark Replay Conditions

Condition	Replay Mode	Mean TSPLIB Gap	Mean Synthetic Gap	Mean Transfer Gap	Code Novelty (mean)	Complexity	Notes on Transfer and Novelty
atsp_no_replay	none	0.1831	0.0090	0.0961	0.885	0.58	Baseline transfer moderate; highest code novelty.
atsp_random_replay	random	**0.1783**	**0.0084**	**0.0933**	0.835	0.78	Best transfer results; slight decrease in code novelty but better gaps.
atsp_failure_replay	failure	0.2100	0.0363	0.1231	0.0	0.78	Poorest transfer and synthetic gaps; zero code novelty.
atsp_failure_replay_compression	failure + compression	0.1937	0.0090	0.1014	0.751	0.58	Moderate transfer; lower code novelty vs no replay. Compression pressure applied.

Interpretation

- **Transfer Performance (TSPLIB & Synthetic Gaps):**

The **random replay** condition (atsp_random_replay) yields the best transfer results on both held-out TSPLIB (~0.178) and synthetic datasets (~0.0084), outperforming no replay and failure replay variants. The improvement is moderate but consistent, indicative of better generalization.

- **Failure Replay Conditions:**

Both failure replay modes yield worse transfer performance, with the no-compression variant showing the highest TSPLIB gap (0.210) and synthetic gap (0.0363), indicating that focusing replay on failure cases does not promote better transfer here. Compression-aware failure replay slightly improves transfer but remains worse than random replay and no replay.

- **Code Novelty vs Transfer:**

Code novelty decreases notably in failure replay conditions, especially with failure replay alone (0), indicating little to no algorithmic exploration or invention. Conversely, random replay reduces novelty slightly compared to no replay (0.835 vs 0.885), but this is accompanied by improved transfer gaps. Thus, **code novelty is not necessarily indicative of better transfer**, and the best-performing condition trades off some novelty for improved optimality.

- **Replay Mode Impact:**
- **No replay ("none"):** moderate transfer, highest code novelty.
- **Random replay:** best transfer, slightly reduced novelty, increased complexity (0.78 vs 0.58).
- **Failure replay:** worst transfer, zero/low novelty.
- **Failure + compression replay:** intermediate transfer and novelty, complexity lower than random replay.
- **Complexity:**

Random replay yields higher complexity (0.78), which might contribute to improved results despite lower novelty. Failure replay variants maintain similar complexity but fail in transfer performance.

Conclusion

- **Random replay enables the best transfer to held-out TSPLIB and synthetic ATSP benchmarks, achieving improved optimality gaps despite a modest reduction in code novelty compared to no replay.**
- **Failure replay strategies degrade transfer performance, with notably zero code novelty in failure replay without compression, indicating little algorithmic innovation.**
- **Compression-aware failure replay somewhat mitigates failure replay's negative impact but remains inferior to random replay in transfer.**
- **The gains in transfer under random replay likely arise from beneficial replay diversity rather than new algorithmic constructs, as code novelty declines somewhat despite improved gaps.**
- **Prioritize random replay for improved generalization to diverse benchmarks, acknowledging a trade-off with code novelty but clear benefits in optimality gaps.**